MA416

Lecture 09 – September 30, 2025
Heteroskedastic ANOVA Testing

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Review on ANOVA Basics

ANOVA Test for Homogeneity

Assumptions...

1. Populations are normally distributed
2. Populations are homoscedastic to one another
3. Independence between samples
4. Independence of observations within samples

We define:

$$\text{SST} = s_{all}^2 \cdot (n-1) \qquad \text{SSE} = \sum_{j=1}^{g} \nu_j \cdot s_j^2 \qquad \text{SSM} = \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{\bar{x}}\right)^2$$

$$MST = \frac{SST}{n-1} \qquad MSE = \frac{SSE}{n-g} \qquad MSM = \frac{SSM}{g-1} \qquad F_{stat} = \frac{MSM}{MSE}$$

Question: What if homoscedasticity assumption (#2) is not met?

Recall the Unpooled Variance ($g = 2$) T-test

$$\mathbb{V}(\bar{X}) \approx \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \qquad \text{and} \qquad t_{stat} = \frac{\bar{x}_d - 0}{\mathrm{SE}(\bar{X}_d)} \sim T_\nu$$

where

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2 \nu_1} + \frac{s_2^4}{n_2^2 \nu_2}}$$

where $0 < \nu \leq n_1 + n_2 - 2 = n - g$.

Reasoning:

- Since populations are homoscedastic, pooled variance (related to SSE) is not appropriate.
- Since different variance leads to higher Type-I error, a more “conservative” degrees of freedom is needed to offset error inflation.

Heteroscedastic ANOVA

We define:

$$w_j = \frac{n_j}{s_j^2} \qquad \text{and} \qquad \bar{\bar{x}}_H = \frac{w_1 \cdot \bar{x}_1 + w_2 \cdot \bar{x}_2 + \cdots + w_g \cdot \bar{x}_g}{w_1 + w_2 + \cdots + w_g}$$

and define:

$$SSM^* = \sum_{j=1}^{g} w_j \left(\bar{x}_j - \bar{\bar{x}}_H\right)^2 \qquad \text{and} \qquad MSM^* = \frac{SSM^*}{g-1}$$

Note: If $s_1^2 = s_2^2 = \cdots = s_g^2 \equiv 1$, *then* $\bar{\bar{x}}_H = \bar{\bar{x}}$ *and* $MSM^* = MSM$.

Heteroscedastic ANOVA

$$w_j = \frac{n_j}{s_j^2} \qquad \text{and} \qquad \bar{\bar{x}}_H = \frac{w_1 \cdot \bar{x}_1 + w_2 \cdot \bar{x}_2 + \cdots + w_g \cdot \bar{x}_g}{w_1 + w_2 + \cdots + w_g}$$

$$SSM^* = \sum_{j=1}^{g} w_j \left(\bar{x}_j - \bar{\bar{x}}_H\right)^2 \qquad \text{and} \qquad MSM^* = \frac{SSM^*}{g-1}$$

Our test statistic will be given by:

$$F^*_{stat} = \frac{MSM^*}{1 + \frac{2(g-2)}{\nu^*}} \qquad \text{where} \qquad \nu^* = \frac{\left(\frac{n_1}{s_1^2} + \frac{n_2}{s_2^2} + \cdots + \frac{n_g}{s_g^2}\right)^2}{\frac{n_1^2}{s_1^4 \nu_1} + \frac{n_2^2}{s_2^4 \nu_2} + \cdots \frac{n_g^2}{s_g^4 \nu_g}}$$

***Question**: Does ν^* equal to the unpooled-variance ν for $g = 2$?*

Heteroscedastic ANOVA

Example: Suppose you plan on conducting an ANOVA test on three different families of data sets. Be able to interpret the associated claim you would be testing, and be able to assess whether the homoscedasticity assumption is met for the following data families.

```
source('mybastat.R') # self-built
# calculates test statistics, p-values, and critical values for Bartlett

mydata1 = read.csv('CompanyPayroll_9.25.csv')
X1 = mydata1$Prior_Experience
Ba1 = mybastat(X1, mydata1$Job_Rank, alpha=0.05)
Ba1$test_stat # approx 1.87
Ba1$pval # approx 0.39

mydata2 = read.csv('EnvStudyData_9.25.csv')
X2 = mydata2$AnnualRainfall
Ba2 = mybastat(X2, mydata2$PollutionLevel, alpha=0.05)
Ba2$test_stat # approx 1.11
Ba2$pval # approx 0.77

mydata3 = read.csv('FlowerGrowth_9.25.csv')
X3 = mydata3$Height_1Month_cm
Ba3 = mybastat(X3, mydata3$Climate_Zone, alpha=0.05)
Ba3$test_stat # approx 19.59
Ba3$pval # approx 0.00006
```

Heteroscedastic ANOVA

Example: Suppose you plan on conducting an ANOVA test on three different families of data sets. Be able to interpret the associated claim you would be testing, and be able to assess whether the homoscedasticity assumption is met for the following data families.

```
# Het. ANOVA Setup
X3s = split(X3, mydata3$Climate_Zone)
nj = sapply(X3s, length)
s2j = sapply(X3s, var)
wj = nj/s2j
xbarbar_h = sum(wj*xbarj)/sum(wj)
ssm_h = sum(wj*(xbarj-xbarbar_h)^2)
ssm_h # approx 200.99

#ssm = sum(nj*(xbarj - mean(X3))^2)
#ssm # approx 20690.77
```

```
g = length(unique(mydata3$Climate_Zone))
msm_h = ssm_h/g
v_h_top = (sum(nj/s2j))^2
v_h_bot = sum(nj^2/(s2j^2*(nj-1)))
v_h = v_h_top/v_h_bot # approx 296.09
n = length(X3) # approx 347

F_stat_h = msm_h/(1 + 2*(g-2)/v_h)
F_stat_h # approx 66.55

pf(F_stat_h, g-1, v_h, lower.tail=F)

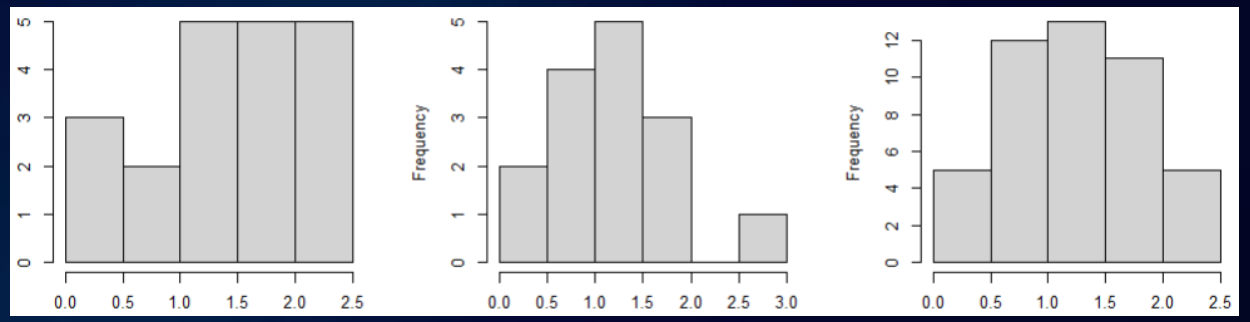
sapply(X3s, mean)
```

```
> sapply(X3s, mean)
     Arid Temperate  Tropical
 10.15779  14.78811  20.41718
```

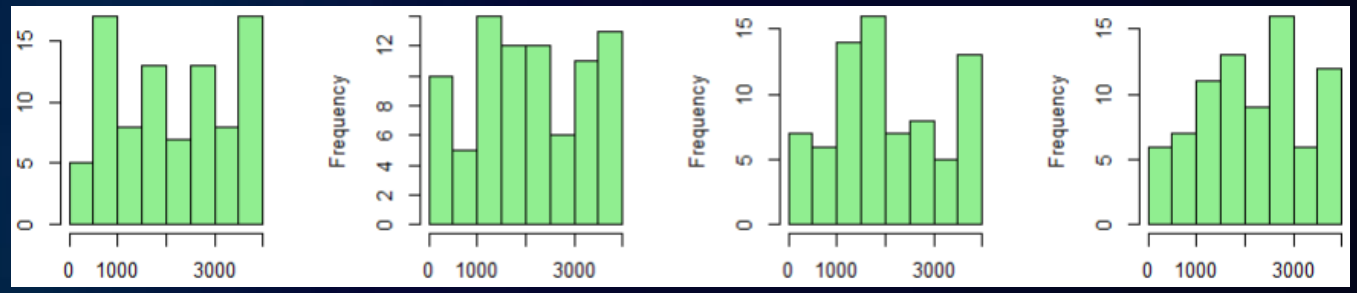
Parametric + Independent ANOVA

It's important to remember that if the normality and/or independence (two-fold) assumptions aren't met, the conclusions of Bartlett and/or ANOVA (either type) can be misleading or moreover wrong.

```
mydata1 = read.csv('CompanyPayroll_9.25.csv')
X1 = mydata1$Prior_Experience
Ba1 = mybastat(X1, mydata1$Job_Rank, alpha=0.05)
Ba1$test_stat # approx 1.87
Ba1$pval # approx 0.39
g1 = length(unique(mydata1$Job_Rank))
X1s = split(X1, mydata1$Job_Rank)
par(mfrow = c(1,g1))
sapply(X1s, hist)
```




```
mydata2 = read.csv('EnvStudyData_9.25.csv')
X2 = mydata2$AnnualRainfall
Ba2 = mybastat(X2, mydata2$PollutionLevel, alpha=0.05)
Ba2$test_stat # approx 1.11
Ba2$pval # approx 0.77
g2 = length(unique(mydata2$PollutionLevel))
X2s = split(X2, mydata2$PollutionLevel)
par(mfrow = c(1,g2))
sapply(X2s, function(x) hist(x, col='lightgreen'))
```




```
mydata3 = read.csv('FlowerGrowth_9.25.csv')
X3 = mydata3$Height_1Month_cm
Ba3 = mybastat(X3, mydata3$Climate_Zone, alpha=0.05)
Ba3$test_stat # approx 19.59
Ba3$pval # approx 0.00006
g3 = length(unique(mydata3$Climate_Zone))
X3s = split(X3, mydata3$Climate_Zone)
par(mfrow = c(1,g3))
sapply(X3s, function(x) hist(x, col='lightpink'))
```

